Economic Complexity Homework II. – Kolos Ágoston Mohácsi, GG9JD0

In the article, Neffke et al. (2017) try to fill two research gaps simultaneously. One is related to Hidalgo et al.'s (2007) work on product space which is cited directly by Neffke et al. (2017), RCA - while innovative - has several gaps in its methodology, notably that it assumes that the co-occurrence of exports on aggregate levels implies a relationship on the individual production factors. Neffke et al.'s (2017) solution is going down to the level of individual cases, analyzing labor flow for the case of human capital. The advantage of this methodology is that by analyzing individual factors of production (human capital in this case), the analysis gives clearer interpretation of which exact components have interchangeability between industries. The second one builds upon this, because while the idea of human capital has an academic consensus within economics, the nature by which human capital can be utilized between activities is debated. Neffke et al.'s (2017) methodology, while agnostic in the paper provides a new framework for measurement on which these theories can be empirically evaluated.

There are two directions of development of Neffke et al.'s (2017) work, one methodological, and one is interpretation related. The methodology development originates from the problem that the proximity between two activities is assumed to be undirected, and thus symmetric for simplicity. Hosseinioun et al. (2025) recently expanded upon this work, by analyzing the US labor flows, while accounting for directions within said flows. Their conclusion is that this asymmetry reveals an important hierarchy in human capital, employment that requires specific human capital usually also requires generalizable human capital, but not vice-versa. The second key problem that arises from Neffke et al.'s (2017) work, is that the model assumes available production capabilities themselves sufficiently justify what an economy produces, and by this implicit assumption the economic coordination is self-regulatory. Frenken et al. (2023) expansion puts an emphasis on not just the capabilities themselves but on the channels for the re-utilization of these capabilities, which „bottleneck" similarly to microeconomics is located within transactional costs. Frenken et al.'s (2023) key conclusion is, thus, that institutional development – by decreasing transactional costs - determines how well labor can be coordinated into complex production.

Both expansions are valid, however what is interesting is the simultaneous usage of these two expansions (which to my knowledge isn't present yet). If complex production requires complex human capital, and complex production only arises if institutions are properly developed, could that mean that the key difference between successful economic specialization and „Dutch Disease" (Corden & Neary, 1982) is found within the institutional effects of human capital accumulation?

Methodology-wise a simple start would be comparative analysis of two nations skill-relatedness networks, which have vastly different institutional frameworks, and examining how similar human capital portfolios yield different structures of production. One good source for this analysis would be the European labor force survey database (Eurostat, n. d.), in which one „Eastern European" and one „Western European" nations labor flows could be compared. Insights from these results could help policy makers understand which aspect of their economy hinders development, institutions, or human capital portfolio.

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AI Usage Statement

Tool used: *Claude (Anthropic), conversational interface*

What I used AI for:

1. Literature search: I asked Claude to help identify citing articles of Neffke et al. (2017) that develop the original work in interesting directions. From the suggestions, I selected Hosseinioun et al. (2025) and Frenken et al. (2023) based on my own reading of these papers.

2. Understanding the human capital debate: I used Claude to map the landscape of competing theories (Becker, Neal, Gathmann & Schönberg, Lazear) to better contextualize Neffke's contribution.

3. Language editing: English is not my first language. I used Claude for grammar checking and iterative language correction across multiple drafts.